

GRAFLY

User Flow Map

A printable reference covering every screen, branch,
and gating rule in the Grafly mobile app.

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App entry & authentication

Cold start runs through the AuthGate, which decides between onboarding, the main app, or the auth screen based on persisted state.

Cold start decision tree

```
Splash / app load
  -> Providers mount: SafeArea, QueryClient, Auth, Game
  -> AuthGate evaluates persisted state

IF onboardingComplete == false
  -> redirect to /onboarding

IF onboardingComplete == true AND route == /onboarding
  -> redirect to /(tabs)

IF route in [auth, auth-callback]
  -> always allowed (password recovery deep link)

ELSE
  -> render requested route
```

Auth screen (/auth) modes

```
[Sign in] (default)
  email + password -> Supabase signIn
    success -> replace /(tabs)
    error    -> inline error
  Continue with Google -> OAuth flow
    success -> replace /(tabs)

[Sign up]
  email + password (>= 6 chars)
    needsConfirmation -> notice + return to Sign in
    confirmed         -> replace /(tabs)

[Reset password]
  email -> Supabase resetPassword
    success -> show 'check your email' notice
    error    -> inline error

Close (X) shown only if onboardingComplete == true -> /(tabs)
```

Onboarding flow

Eight ordered steps from welcome to optional account creation. Placement test can be skipped; account creation can be deferred.

Step sequence

```
1. welcome
  - Logo, theme toggle (light/dark), Start CTA
  v
2. name
  - TextInput (max 24 chars), required to advance
  v
3. goal
  - Pick 1 of 4: basics | improve | portfolio | career
  v
4. level (self assessed)
  - Beginner | Intermediate | Advanced
  v
5. time (daily commitment)
  - 5 min | 10 min | 15+ min
  v
6. placement
  - Multi choice and true/false questions
  - 'Skip placement' -> jump to results as 'novice'
  v
7. results
  - PlacementLevel: novice | beginner | intermediate | advanced | expert
  - Mascot celebrates
  v
8. signup
  - Continue with Google -> finish
  - Email sign in / sign up / reset -> finish
  - 'Skip / continue without account' -> finish (guest)
  v
completeOnboarding(name, level) -> redirect /(tabs)
```

Placement scoring

```
score / total  
>= 0.90 -> expert  
>= 0.75 -> advanced  
>= 0.55 -> intermediate  
>= 0.35 -> beginner  
< 0.35 -> novice
```

Main tab navigation overview

After onboarding, users land in a 5 tab floating bottom nav. Each tab is a top level screen; lessons, paywall, leaderboard, and auth are presented above it.

Tab bar (left to right)

```
[ Home ]    index    -> /(tabs)/index
[ Tree ]    tree     -> /(tabs)/tree
[ Critique ] critique -> /(tabs)/critique
[ Shop ]    shop     -> /(tabs)/shop
[ Profile ] profile  -> /(tabs)/profile
```

Modal / above tab routes

```
/lesson?nodeId=... (full screen lesson player)
/paywall            (subscription upsell)
/leaderboard        (weekly XP leaderboard)
/courses            (course library)
/auth               (sign in / up / reset)
/auth-callback      (OAuth + password reset return)
```

Home tab flow

Daily landing page summarising progress, surfacing the next lesson, and routing to courses, the leaderboard, and the paywall.

Sections (top to bottom)

```
Header (fixed, minimal)
  - Logo, streak count (flame icon), coin count

Greeting + streak status
  - 'Hey {name}', current streak, streak risk warning

Pro upgrade banner (free users only)
  -> tap -> /paywall

Continue learning card
  -> tap -> /lesson?nodeId=<lastUnfinished>

Popular courses carousel
  -> tap card -> /(tabs)/tree?courseId=<id>

Rank / leaderboard card
  -> tap -> /leaderboard

Overlays
  - XPPopup on XP gain
  - LevelUpOverlay on level up
```

Skill Tree tab flow

Course picker plus a vertical node tree. Nodes have three states (locked / unlocked / completed); pressing one opens a sheet that launches the lesson.

Course picker

```
Header shows current course + completion %
-> tap header -> course picker modal
    -> select course -> tree re renders for that course
```

Node states

```
For each node:
  IF every prerequisite node id is in completedLessons
    IF every lesson in node is completed
      -> COMPLETED (success colour, checkmark)
    ELSE
      -> UNLOCKED (course colour, lesson icon)
  ELSE
    -> LOCKED (muted colour, lock icon)
```

Node press

```
Tap node -> NodeSheet (bottom sheet)
- Title, description, lesson count, XP reward
IF locked -> message 'Complete previous lessons' (no CTA)
IF unlocked -> 'Start lesson' -> /lesson?nodeId=<id>
IF completed -> 'Review' -> /lesson?nodeId=<id>
```


Lesson flow

Question by question lesson player. Seven question types share one engine. Wrong answers cost a heart; reaching zero hearts forces the paywall.

Per question loop

Render question by type:

- multiple_choice
- true_false
- spot_the_difference
- tap_the_element
- arrange_in_order
- drag_to_match
- fill_in_blank

User submits answer

IF correct

- > success colours, checkmark, success sound
- > mascot 'celebrate'
- > tap Continue -> next question

IF wrong

- > destructive colours, shake, error sound
- > mascot 'wrong'
- > loseHeart()

IF hearts == 0

- > redirect /paywall (out of hearts state)

ELSE

- > tap Continue -> next question

End of lesson summary

After last question:

- completeLesson(nodeId, lessonIdx)
- Award XP, coins; record completion
- Show summary: XP earned, coins earned, hearts lost

Summary CTAs:

IF more lessons in this node

- > 'Next Lesson' -> advance lessonIdx, reset state

ELSE (last lesson in node)

- > CTA -> router.replace('/:tabs)/tree')

Critique tab flow

AI design mentor. Pick a mockup, chat in turns, earn rewards after enough exchanges. Free users are capped per day; Pro is unlimited.

Session lifecycle

1. Idle / mockup selection
 - Shuffle mockup deck, tap one to start
 - ▼
2. Mentor chat
 - User submits critique notes
 - AI responds (typewriter animation)
 - Repeat for several turns
 - ▼
3. Reward gate
 - IF userTurns >= MIN_USER_TURNS_FOR_REWARD (3)
 - > award XP + coins, mark session complete
 - ELSE
 - > no reward yet, can keep chatting
 - ▼
4. Session end
 - Reset to idle, show 'sessions remaining today'

Free vs Pro gating

- Free tier
- maxSessions = 2 (per screen mount, local useState)
 - sessionsDone resets if the user leaves and returns
 - On limitReached:
 - > show 'Limit reached' state
 - > CTA 'Unlock Pro' -> /paywall
- Pro tier
- maxSessions = Infinity (state.isPro == true)
 - No paywall trigger from this screen

Shop tab flow

Browse power ups and cosmetics. Power ups deduct coins and persist to game state; cosmetics show a confirmation toast only. Insufficient balance shows a friendly fail state.

Categories

Power ups (persistent)

- Heart Refill -> refillHearts() (costs 100 coins)
- Streak Shield -> purchaseShield()
- XP Booster -> activateBooster()

Cosmetics (visual only)

- Avatar Frames -> 'equipped!' toast (no coin deduction, no persisted equip state in current build)

Purchase flow

Tap item -> ShopCard checks `canAfford = state.coins >= cost`

Power up branch

```
IF canAfford
  -> useCoins(cost) + apply effect via GameContext
  -> mascot 'celebrate' + success toast
ELSE
  -> mascot 'wrong' + 'Not enough coins' toast
```

Cosmetic branch

```
IF canAfford -> 'X equipped!' toast (UI only)
ELSE          -> 'Not enough coins' toast
```

No Pro only gating in current shop build

Profile tab flow

User identity, gamification stats, achievements, app preferences, and account actions in one scroll.

Sections (top to bottom)

Identity card

- Mascot, username (tap pencil to edit)
- Level, division (bronze..diamond), placement label
- Weekly XP

Progress

- Level number, XP bar (current / required)
- 'X XP to level N+1'

By the numbers (stats grid)

- Total XP, current streak, max streak, lessons done

Achievements / badges (horizontal scroll)

- Unlocked badges in colour, locked in muted with lock icon

Preferences

- Voice feedback toggle (toggleVoice)

Account

- Upgrade to Pro -> /paywall
- IF signed in -> Sign out (signOut)
- IF guest -> 'Save your progress' -> /auth

Paywall flow

Single paywall screen reached from many trigger points. Annual plan is preselected; purchase flips isPro and returns the user to where they came from.

Trigger points

```
Home upgrade banner
Critique 'limit reached' CTA
Lesson hearts == 0 (forced)
Profile 'Upgrade to Pro' CTA
Shop Pro only item tap
```

Plans + purchase

```
Plan picker
- Annual $39.99 / yr (preselected, 'SAVE 33%' badge)
- Monthly $4.99 / mo ('Cancel anytime')
```

```
Sticky CTA
-> handlePurchase()
  - simulate processing (~1.2s)
  - dispatch SET_PRO { isPro: true }
  - router.back() (or replace '/')
```

```
Close (X) -> back / replace '/'
```

Cross cutting states

Behaviours that span tabs and screens. All flow through GameContext / AuthContext, persisted to AsyncStorage and (if signed in) Supabase.

Hearts

```
Wrong answer -> loseHeart()
Hearts only refill via explicit actions:
  - Shop 'Heart Refill' (REFILL_HEARTS, costs 100 coins)
  - No automatic time based refill in current build
IF hearts == 0 mid lesson -> /paywall (out of hearts)
```

Streak

```
completeLesson() runs INCREMENT_STREAK
isConsecutive = (lastStreakDate == yesterday.toDateString())
IF isConsecutive -> streak += 1
ELSE -> streak resets to 1
streakMax = max(streakMax, newStreak)

Streak Shield (Shop) increments shield count on purchase,
but is not auto consumed when a day is missed in current build.
```

Level up

```
XP awarded by lesson + critique reward
IF xp crosses threshold for next level
  -> level += 1
  -> LevelUpOverlay shown on Home / current screen
  -> mascot 'celebrate'
```

Badges / achievements

Each badge has a condition(state) -> bool

Examples:

- First Step : completedLessons.length >= 1
- On Fire : streakMax >= 7
- Rising : level >= 5
- Pro : level >= 10
- Rich : coins >= 500
- Dedicated : completedLessons.length >= 10

Recomputed each Profile render; locked ones dimmed

Sync

GameContext writes to AsyncStorage on every change

IF authenticated

- > debounced (1.5s) push to Supabase
- > on app open: hydrate from Supabase, fall back to local

IF guest

- > local only; sign up later migrates local state